🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 8 · STANDARD 6.AT.8

Write and Solve Equations Using Multiplication or Division

Lesson 8-3 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes

Level 3 Enrichment

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can solve one-step multiplication and division equations using inverse operations.
Language Objective: I can explain my steps using the words inverse operation, multiply, divide, and isolate.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Solve Multiplication and Division Equations | Inverse operation | Multiply | Divide | Isolate | Coefficient | Solution

👆 *Tap any word to see what it means and a picture.*

Level 3: try the blanks from memory first, then check the bank.

1. Undo a multiplication by ________ both sides by the same number.

2. An operation that undoes another, like division undoing multiplication, is an ________.

3. To undo dividing by a number, I ________ both sides by that number.

4. To undo multiplying by a number, I ________ both sides by that number.

5. To get the variable alone on one side is to ______ it.

6. The number multiplied by a variable is the ______.

7. The value of the variable that makes an equation true is the ______.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

How many kits did the detective order? How did inverse operations help you solve this?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- [ ] **Solve Multiplication and Division Equations** — Use inverse multiplication or division to isolate the variable.
  Resolver ecuaciones de multiplicación y división — Usar la multiplicación o división inversa para aislar la variable.
- [ ] **Inverse operation** — Two math actions that undo each other, like × and ÷.
  Operación inversa — Dos operaciones que se deshacen entre sí, como × y ÷.
- [ ] **Multiply** — To add equal groups again and again.
  Multiplicar — Sumar grupos iguales una y otra vez.
- [ ] **Divide** — To split into equal groups.
  Dividir — Repartir en grupos iguales.
- [ ] **Isolate** — To get the letter by itself on one side.
  Despejar — Dejar la letra sola en un lado.

Say it first: Say your idea to a partner or quietly to yourself before you write.
Di tu idea a un compañero o en voz baja antes de escribir.

The detective ordered ____ kits because 7k = 63 means k = 63 ÷ ____ = ____.
Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: In 3x = 21 the variable is multiplied, so I undo it with division.

Evidence: Students read 3x as 3 groups of x, recognize multiplication, and contrast it with addition, knowing they will divide (not subtract) to solve.

Reasoning: This evidence connects the definition of solve multiplication and division equations to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The detective ordered ____ kits because 7k = 63 means k = 63 ÷ ____ = ____.**
- **My answer is ____.**
  Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **I know because ____.**
  Lo sé porque ____.
- **So ____.**
  Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
  Mi afirmación es ____.
- **My math evidence is ____.**
  Mi evidencia matemática es ____.
- **This proves ____ because ____.**
  Esto demuestra ____ porque ____.

4. Check Your Explanation

- [ ] I answered the question.
  Respondí la pregunta.
- [ ] I used math evidence: numbers, an equation, a model, or a comparison.
  Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- [ ] I explained how the evidence proves my answer.
  Expliqué cómo la evidencia demuestra mi respuesta.
- [ ] I used at least two math words.
  Usé por lo menos dos palabras matemáticas.
- [ ] I reread complete sentences and punctuation.
  Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Solve Multiplication and Division Equations
2. **Notes 2:** Inverse operation
3. **Notes 3:** Multiply
4. **Notes 4:** Divide
5. **Notes 5:** Isolate
6. **Notes 6:** Coefficient
7. **Notes 7:** Solution
8. **Watch for this mistake:** *A common mistake in Solve Multiplication and Division Equations is applying the SAME operation again instead of the inverse — for example, seeing 6x = 42 and multiplying both sides by 6 to get x = 252, instead of dividing by 6 to get x = 7. Before you submit, ask: did I use the OPPOSITE operation shown in the equation, and did I apply it to both sides?*
9. **Try It 1:** A. x = 18 *— x is divided by 3, so multiply both sides by 3: x = 6 x 3 = 18. Check: 18 / 3 = 6.*
10. **Try It 2:** A. n = 9 *— The 5 is multiplying n, so divide both sides by 5: n = 45 / 5 = 9. Check: 5 x 9 = 45.*